

CELNAV quick reference

Menu: run celnav. Everything also works from the command line.

celnav dr "35 10.4 N" "040 20.1 W" 245 11	set DR, course true, speed
celnav plan	twilight, what is up, best three
celnav sight "2026-08-29 07:30:00" Dubhe C "19 32.1" 1.5 3.0	UT, body, limb, Hs, index error, eye height
celnav fix	reduce the log and plot the fix
celnav compass "UT" sun 112.5 6.0	compass error and deviation
celnav log / celnav clear	list / empty the sight log
celnav doctor	environment check and self test
celnav learn	lessons, walkthrough, drills, sandbox
celnav drill red	one marked drill (corr, alm, red, int, full, fix)
celnav night / day / plain	colour mode

Menu keys: 1 sight · 2 fix · 3 plan · 4 almanac · 5 compass · 6 DR · 7 log · 8 settings · 9 learn · s stars · t test · d check · c colour · h help · q quit

Order of work

1. **DR** — position, course made good, speed. Without course and speed the sights are not advanced.
2. **Plan** — before twilight. Note the three bodies and their approximate altitudes so you can find them fast.
3. **Shoot** — record UT to the second. *Four seconds is one mile of longitude.*
4. **Enter** each sight. Limb: **L** lower, **U** upper, **C** centre (stars and planets).
5. **Fix** — read the residuals and the geometry line before you use it.
6. **Clear** the log.

Reading the answer

- **Intercept** — miles from your DR toward (+) or away (–) from the body.
- **Residual** — how far each line of position misses the fix. This is your observing error. Under a mile is good work.
- **RMS scatter** — the same thing for the round as a whole.
- **GEOMETRY** — how far one arcminute of sight error moves the fix. Under 1.6 nm is a sound cut; over 3 nm is a weak one and CELNAV says so.

Formats

Angles	35 10.4 N	040 20.1 W	-35.1733	040:20.1W
Altitude	32 14.6	(degrees and decimal minutes)		
Time	2026-08-29 07:30:00 UT (or 073000 in the menu, today's date assumed)			

The screen

The position the sights are reduced from.

The sextant and weather figures now in force.

The working cycle: enter each sight as you take it, then reduce the whole round at once.

Run before twilight - what is up, how high, and the best three to shoot.

Type a number and press return.

Checks your awk, your clock and the data folder, then runs the built-in self test.

Course made good and speed. With these set, every line of position is advanced to the fix time for you.

Sights waiting to be reduced.

Sets the two figures highlighted above.

CELNAV 1.0 celestial navigation, no internet needed

DR 35 00 N 040 00 W course 000 T speed 0 kn

eye 2.5 m index error 0.0' 10 C 1010 mb sights logged: 3

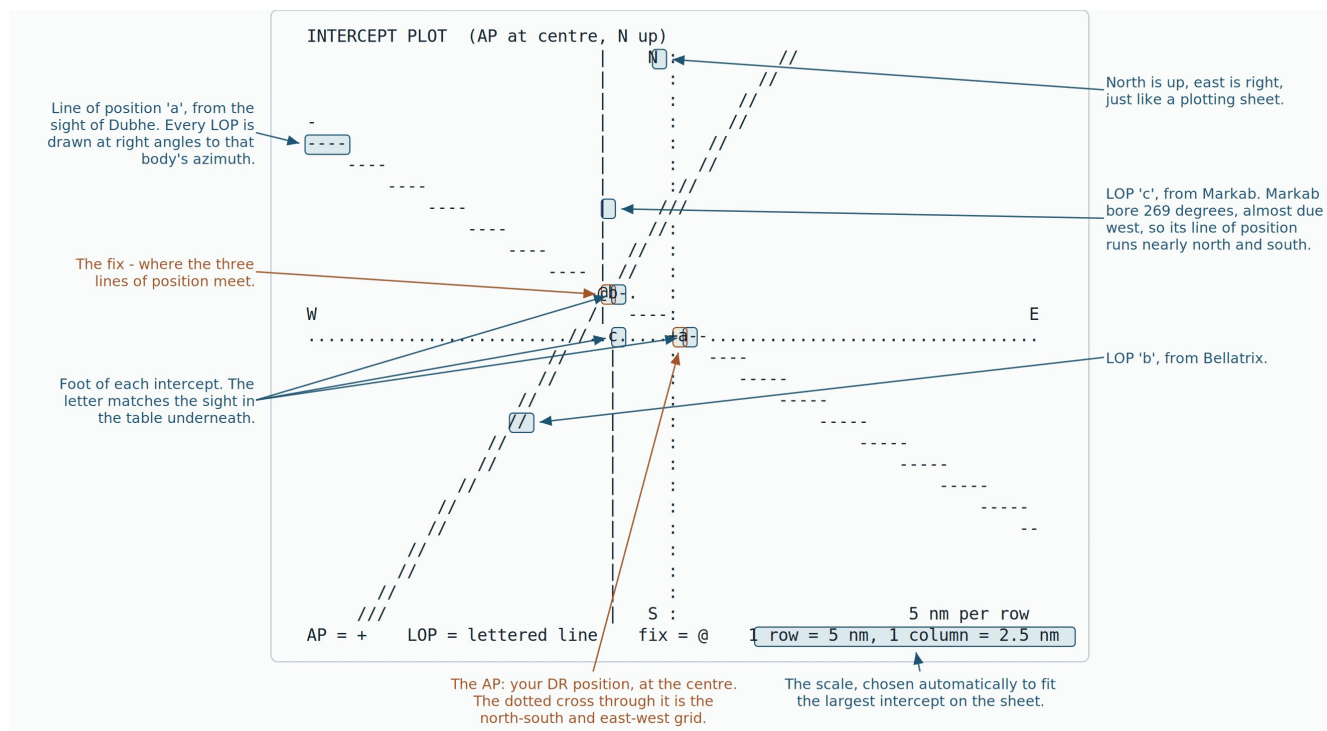
1 Enter a sight 2 Reduce sights -> FIX 3 Sight planning / stars 4 Almanac for a time

5 Compass check 6 Set DR, course and speed 7 Sight log 8 Sextant and weather settings

s Star list t Self test d Check setup h Help q Quit

> 2

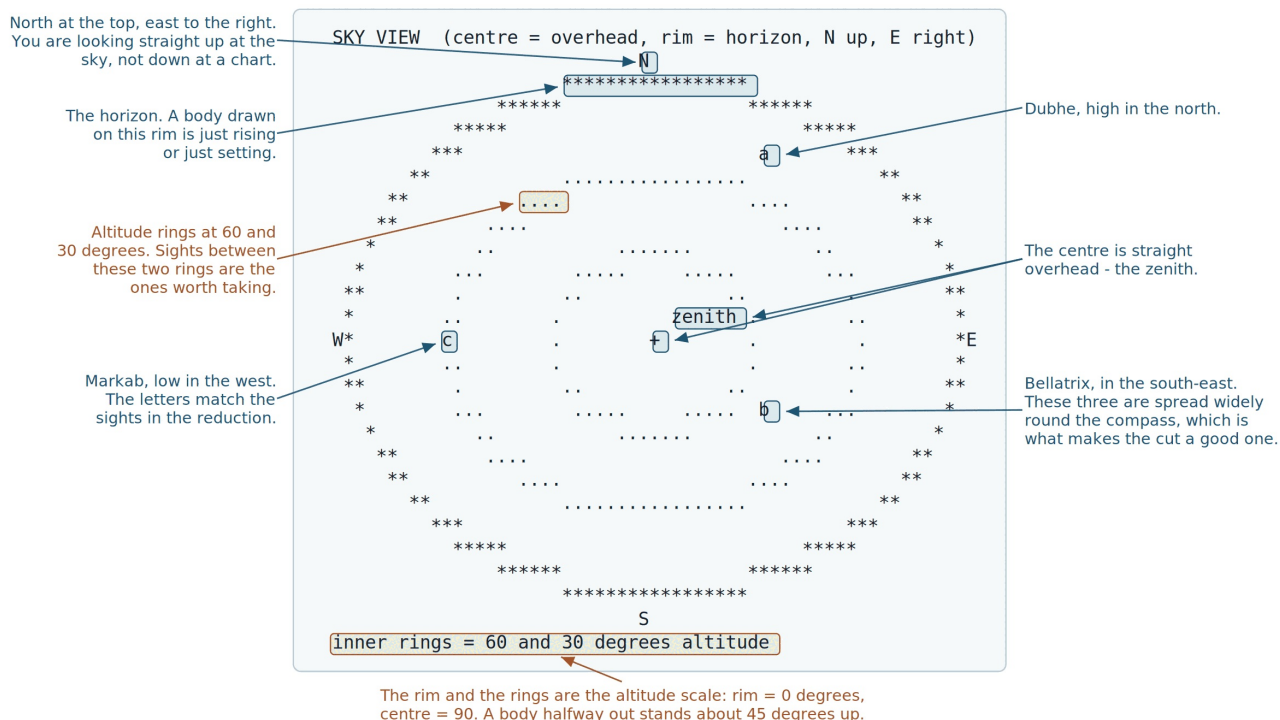
The menu. The two status lines always show what the next reduction will be based on.



The intercept plot. Your DR at the centre, one lettered line for each sight, the fix where they cross.

Reading the plot: + is your DR at the centre of the sheet · each lettered line is a line of position, drawn at right angles to that body's azimuth · the letter sits at the foot of its intercept · @ is the fix · the scale is printed at the bottom right and is chosen to fit the largest intercept.

The sky view



The sky view from `celnav plan` — centre is straight overhead, the rim is the horizon, north is up. Bodies spread widely round the compass give the tightest fix.

When something is wrong

Symptom	First thing to check
Fix far from DR with large residuals	the time of each sight — a wrong digit in the seconds moves longitude only
“Single line of position only”	two sights on nearly the same azimuth — add a body 40–120° away
“no awk with trigonometric functions”	<code>apk add gawk</code> on iSH, <code>pkg install gawk</code> on Termux
Self test fails	the file transfer truncated — check <code>cksum celnav</code>

Colour

`c` in the menu, or `celnav day / night / plain`.

- **day** — white on black, highlights in green.
- **night** — deep red on black, highlights in bright red. No green at night: twenty minutes of dark adaptation is not worth a prettier screen.
- **plain** — no escape sequences; chosen automatically when output is not a terminal.

Learning it

`celnav learn`, or menu item 9.

- **Lessons** — twenty, in four modules, from what a sight measures to what makes a cut weak. Each ends with a question; answer it and the lesson is ticked off.
- **Walkthrough** — one real sight in ten steps, every line explained as it is computed.
- **Drills** — problems built from the real almanac and marked, with the full working shown afterwards. Types: `corr`, `alm`, `red`, `int`, `full`, `fix`.
- **Sandbox** — change latitude, declination or LHA and watch the navigational triangle move.

Rules of thumb

- Prefer bodies between **15° and 70°** of altitude.
- Three bodies about **120° apart** in azimuth give the tightest fix.

- The **star horizon** is usable between civil and nautical twilight — a window of roughly twenty minutes.
- A sight of the Sun at any altitude gives a line; two of them, an hour or more apart, give a running fix.